

Surname _____

Van _____

Initials _____

Voorletters _____

Stud. no. _____

The undersigned hereby confirms that he/she is familiar with and accepts the examination regulations

Ondergetekende bevestig hiermee dat hy/sy op hoogte is met die eksamenregulasies en dat hy/sy hom/haar daaraan onderwerp

Handtekening _____ Signature _____ Date _____ Datum _____

$v_{avg} = \frac{1}{2}(v_0 + v)$	$v^2 = v_0^2 + 2a(x - x_0)$	$a = \frac{v^2}{r}$	$R = \frac{v_0^2}{g} \sin 2\theta_0$
$v = v_0 + at$	$x - x_0 = \frac{1}{2}(v_0 + v)t$	$g = 9,8 \text{ m/s}^2$	$T = \frac{2v_0 \sin \theta}{g}$
$x = x_0 + v_0 t + \frac{1}{2}at^2$	$x - x_0 = vt - \frac{1}{2}at^2$	$h = \frac{v_0^2 \sin^2 \theta}{2g}$	
$x = x_0 + v_{avg} t$			

Question 1/Vraag 1 ch 7:Q2

Figure (a) shows two horizontal forces that act on a block that is sliding to the right across a frictionless floor. Figure (b) shows three plots of the block's kinetic energy K vs time t . Which of the plots best correspond to the following three situations:

Figuur (a) dui twee horisontale kragte aan wat inwerk op 'n blok wat na regs oor 'n wrywinglose vloer gly. Figuur (b) dui drie grafieke van die kinetiese energie K teenoor tyd t aan.

Watter een van die grafieke stem ooreen met die volgende situasies:

(a) $F_1 = F_2$?

[2]

2

(b) $F_1 > F_2$?

[2]

3

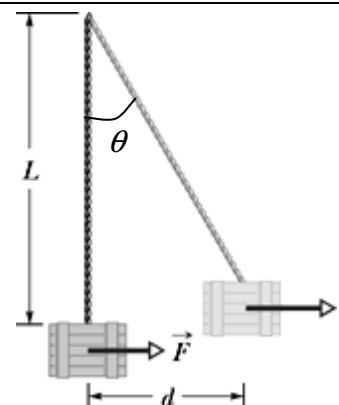
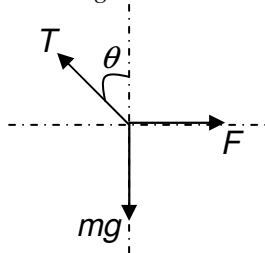
Question 2/Vraag 2 ch 7:P2 homework

A 230 kg crate hangs from the end of a rope of length $L = 12.0 \text{ m}$. You push horizontally on the crate with a varying force $\vec{F}$ to move it a distance $d = 4.00 \text{ m}$ to the side (See the figure).

'n 230 kg krat hang aan 'n tou met lengte $L = 12.0 \text{ m}$. Jy stoot die krat horisontaal met 'n variërende krag, $\vec{F}$ en dit beweeg 'n afstand $d = 4.00 \text{ m}$ na die kant toe (Sien die figuur).

(a) Determine the magnitude of $\vec{F}$ when the crate is at rest in this final position.

Bepaal die grootte van $\vec{F}$ wanneer die krat met rus is by dié finale posisie.



[2+5]

Determine the angle the rope makes with the vertical. $\theta = \sin^{-1}\left(\frac{4.00}{12.0}\right) = 19.5^\circ$.

Apply Newton's 2nd law to the crate:

$$\sum F_y = ma_y = 0 = T \cos \theta - mg$$

$$\therefore T = \frac{mg}{\cos \theta}$$

$$(T = 2391 \text{ N})$$

$$\sum F_x = ma_x = 0 = F - T \sin \theta$$

$$F = T \sin \theta = \left(\frac{mg}{\cos \theta}\right) \sin \theta$$

$$F = mg \tan \theta$$

$$F = 797 \text{ N}$$

During the crate's displacement, determine the following:

Gedurende die krat se verplasing, bepaal die volgende:

(b) The total work done on the crate.

Die totale arbeid verrig op die krat.

[3]

The crate is initially at rest, and its final position is also at rest

Therefore there is no change in kinetic energy, and according to kinetic energy-work theorem the net work on it is zero.

(c) The work done by the gravitational force on the crate.

Die arbeid verrig deur die swaartekrag op die krat.

[3]

Determine the change in height of the crate:

In final position – hypotenuse of triangle is 12.0 m.

From pythagorous:

$$\text{The vertical side is } L' = \sqrt{L^2 - d^2} = \sqrt{12^2 - 4^2} = 11.31 \text{ m}$$

$$\text{Or Trig: } h = L(1 - \cos \theta)$$

Change in vertical position $h = +0.687 \text{ m}$ (upwards)

The work done by gravity is $W_g = \vec{F}_g \cdot \vec{d} = -mgh$, (angle between $\mathbf{F}_g$ and $\mathbf{d}$ is 180°)

we obtain $W_g = -1547 \text{ J}$ which should be rounded to three figures: -1.55 kJ .

(d) The work done by the pull on the crate from the rope?

Die arbeid verrig deur die tou wat die krat optrek.

[3]

The tension vector is everywhere perpendicular to the direction of motion, so its work is zero (since $\cos 90^\circ = 0$).

(e) Knowing that the crate is motionless before and after its displacement, use the answers to (b), (c), and (d) to find the work your force does on the crate.

Met die kennis dat die krat voor en na die verplasing nie beweeg nie, gebruik die antwoorde van (b), (c) en (d) om te bepaal wat die arbeid is wat verrig word deur jou krag op die krat.

[3]

From (b) $W_{net} = 0 \text{ J}$

But: $W_{net} = W_T + W_{mg} + W_F$

Therefore:

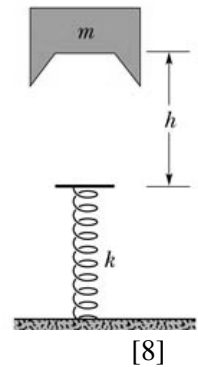
$$W_{net} = 0 + W_{mg} + W_F$$

Thus, $W_F = -W_g = 1.55 \text{ kJ}$.

Question 3/Vraag 3 ch 8:P26

A block of mass $m = 2.0 \text{ kg}$ is dropped from a height $h = 40 \text{ cm}$ onto a spring of spring constant $k = 1960 \text{ N/m}$ (See the figure). Find the maximum distance the spring is compressed.

'n Blok met massa $m = 2.0 \text{ kg}$ val vanaf 'n hoogte $h = 40 \text{ cm}$ op 'n veer met die veerkonstante $k = 1960 \text{ N/m}$ (Sien die figuur). Bepaal die maksimum afstand waardeur die veer saamgepers word.



[8]

Reference points for PE:

Initial position of block for gravitational PE of Block is zero gravitational potential energy:

Relaxed position of spring for spring PE: $x = 0$

$m = 2.0 \text{ kg}$

$h = 0.40 \text{ m}$

x denotes the compression of the spring

The block drops a total distance $h + x$,

Initial PE of block:

$U_{i, \text{block}} = 0 \text{ J}$

$U_{i, \text{spring}} = 0 \text{ J}$

Final PE of block:

$U_{f, \text{block}} = -mg(h+x)$

$U_{f, \text{spring}} = \frac{1}{2} kx^2$

kinetic energy of the block is zero both at the initial and final positions.

Since energy is conserved

$$K_i + U_i = K_f + U_f$$

$$0 = -mg(h+x) + \frac{1}{2} kx^2$$

Using this quadratic formula, its solution is

$$x = \frac{mg \pm \sqrt{(mg)^2 + 2mghk}}{k}.$$

Now $mg = 19.6 \text{ N}$, $h = 0.40 \text{ m}$, and $k = 1960 \text{ N/m}$, and we choose the positive root giving $x = 0.099 \text{ m}$.

Note: Other reference points for zero potential energy can be taken too – e.g. on the top of the spring, or at the max. compression of the spring.

THE END / DIE EINDE